

Problem Statement 1

AI-Generated Voice Detection (Tamil, English, Hindi, Malayalam, Telugu)

1. Introduction

AI systems can now generate very realistic human-like voices. Because of this, it is difficult to identify whether a voice recording was spoken by a real human or generated by an AI system.

In this problem, students must build an API-based solution that detects whether a given voice sample is AI-generated or Human, across five supported languages.

2. Supported Languages (Fixed)

Your system must support only these five languages:

- Tamil
- English
- Hindi
- Malayalam
- Telugu

Each request will contain one audio file in one of the above languages.

3. What You Need to Build

You must design and deploy a REST API that:

- Accepts one MP3 audio file at a time
- Audio will be sent only as Base64
- Analyzes the voice
- Returns whether the voice is:
 - **AI_GENERATED**
 - **HUMAN**
- Responds in JSON format
- Is protected using an API Key

4. Input Rules

- Audio format: MP3
- Input type: Base64 encoded
- One audio per request
- Audio must not be modified

5. API Authentication

Your API must validate an API Key sent in request headers.

API Key Header Format

- **x-api-key:** YOUR_SECRET_API_KEY

Requests without a valid API key must be rejected.

6. API Request (cURL Example)

Endpoint Example

- POST https://your-domain.com/api/voice-detection

cURL Request Example

```
curl -X POST https://your-domain.com/api/voice-detection \
```

```
-H "Content-Type: application/json" \
```

```
-H "x-api-key: sk_test_123456789" \
```

```
-d '{
```

```
  "language": "Tamil",
```

```
  "audioFormat": "mp3",
```

```
  "audioBase64": "SUQzBAAAAAAAAI1RTU0UAAAAPAAADTGF2ZjU2LjM2LjEwMAAAAAAAAA..."
```

```
}'
```

7. Request Body Fields

Field	Description
language	Tamil / English / Hindi / Malayalam / Telugu
audioFormat	Always mp3
audioBase64	Base64-encoded MP3 audio

8. API Response Body (Success)

Example Response

- {
- "status": "success",
- "language": "Tamil",
- "classification": "AI_GENERATED",
- "confidenceScore": 0.91, // out of 1.0
- "explanation": "Unnatural pitch consistency and robotic speech patterns detected"
- }

9. Response Field Explanation

Field	Meaning
-------	---------

status	success or error
language	Language of the audio
classification	AI_GENERATED or HUMAN
confidenceScore	Value between 0.0 and 1.0
explanation	Short reason for the decision

10. Classification Rules (Strict)

Only one classification field is required:

- AI_GENERATED → Voice created using AI or synthetic systems
- HUMAN → Voice spoken by a real human

👉 voiceSource is removed because it is logically the same as classification.

11. Error Response Example

```
{  
  
  "status": "error",  
  
  "message": "Invalid API key or malformed request"  
}
```

12. Evaluation Process

1. System sends one Base64 MP3 per request

2. Language will be one of the 5 supported languages
3. Your API analyzes the voice
4. JSON response is returned
5. Multiple requests are made for evaluation

13. Evaluation Criteria

Participants will be evaluated on:

- 🎯 Accuracy of AI vs Human detection
- 🌐 Consistency across all 5 languages
- 📦 Correct request & response format
- ⚡ API reliability and response time
- 🧠 Quality of explanation

14. Rules & Constraints

- ❌ Hard-coding results is strictly prohibited
- ❌ Misuse of data leads to disqualification
- ⚠️ External detection APIs may be restricted
- ✅ Ethical and transparent AI usage is mandatory

15. One-Line Summary

Build a secure REST API that accepts one Base64-encoded MP3 voice in Tamil, English, Hindi, Malayalam, or Telugu and correctly identifies whether it is AI-generated or Human.

16. Sample Reference Voice:

Drive link - [📁 sample voice 1.mp3](#)